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React JS Lesson 1 – College Coders Notes

What is React?

- React is a JavaScript library used to build user interfaces.
- It is mainly used for creating dynamic web applications.
- React allows developers to build reusable UI parts called components.
- It was developed by Facebook.

Key Features of React

- Component-based architecture
- Virtual DOM for faster updates
- Reusable code structure
- One-way data flow
- JSX support

Why React is Fast – Virtual DOM vs Real DOM

- Real DOM updates the entire page structure directly.
- Virtual DOM creates a lightweight copy of the real DOM.
- React compares old and new Virtual DOM and updates only changed parts.

Why React is a Single Page Application (SPA)

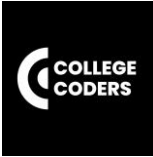
- SPA loads one HTML page once.
- Only content changes without full page reload.
- Example: moving between Home and About page without browser refresh.

React Project Setup (Vite / CRA)

- Vite setup: `npm create vite@latest`
- CRA setup: `npx create-react-app myapp`
- Install dependencies using `npm install`
- Run project using `npm run dev`

Folder Structure & File Usage Explained

- `src` folder contains main React code
- `components` folder stores reusable components
- `App.jsx` is root component
- `main.jsx` starts React app



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What are Components? – Functional Component Explanation

- Components are reusable UI blocks.
- Functional components are JavaScript functions returning JSX.

Portfolio Project Structure Setup

- Create folders: assets, components, pages
- Separate Navbar, Hero, Projects, Contact sections

Rendering, Re-rendering, and How React Updates UI Faster

- Rendering means displaying UI first time.
- Re-rendering happens when state changes.
- React updates only changed parts using Virtual DOM.

Code Example:

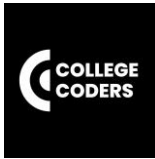
```
function App(){  
  return <h1>Hello React</h1>  
}
```

Explanation:

- App is a functional component.
- return sends JSX to browser.
- h1 is rendered on screen.

Detailed Project Setup Steps

- Step 1: Install Node.js from official website before starting React.
- Step 2: Open terminal and run: `npm create vite@latest`
- Step 3: Enter project name and choose React framework.
- Step 4: Select JavaScript as variant.
- Step 5: Move into project folder using `cd project-name`.
- Step 6: Install packages using `npm install`.
- Step 7: Start project using `npm run dev`.
- Step 8: Open localhost URL in browser.



React Project Folder Explanation

- `node_modules` → stores installed packages automatically.
- `public` → stores static files like images and icons.
- `src` → main working folder for React development.
- `package.json` → project information and dependencies.
- `vite.config.js` → Vite configuration file.

Inside src Folder Files

- `main.jsx` → React application entry point.
- `App.jsx` → main root component loaded first.
- `index.css` → global styles applied to project.
- `components` folder → reusable UI sections.

Portfolio Project Recommended Structure

- `components/Navbar.jsx` → navigation bar.
- `components/Hero.jsx` → homepage top section.
- `components/About.jsx` → profile section.
- `components/Projects.jsx` → project showcase.
- `components/Contact.jsx` → contact form.